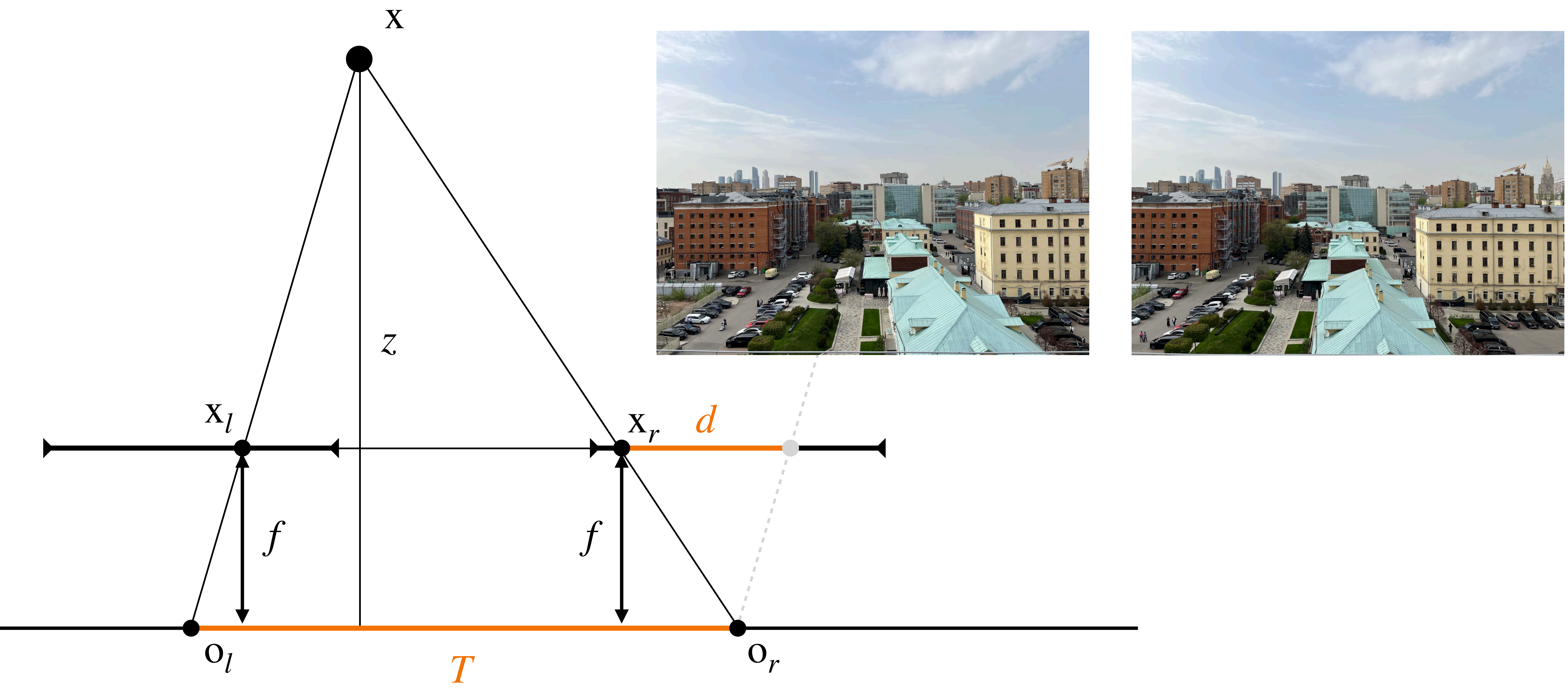
IV: Recent Advances in Multi-view Transformers

3D CV

Kirill Struminsky, Mishan Aliev

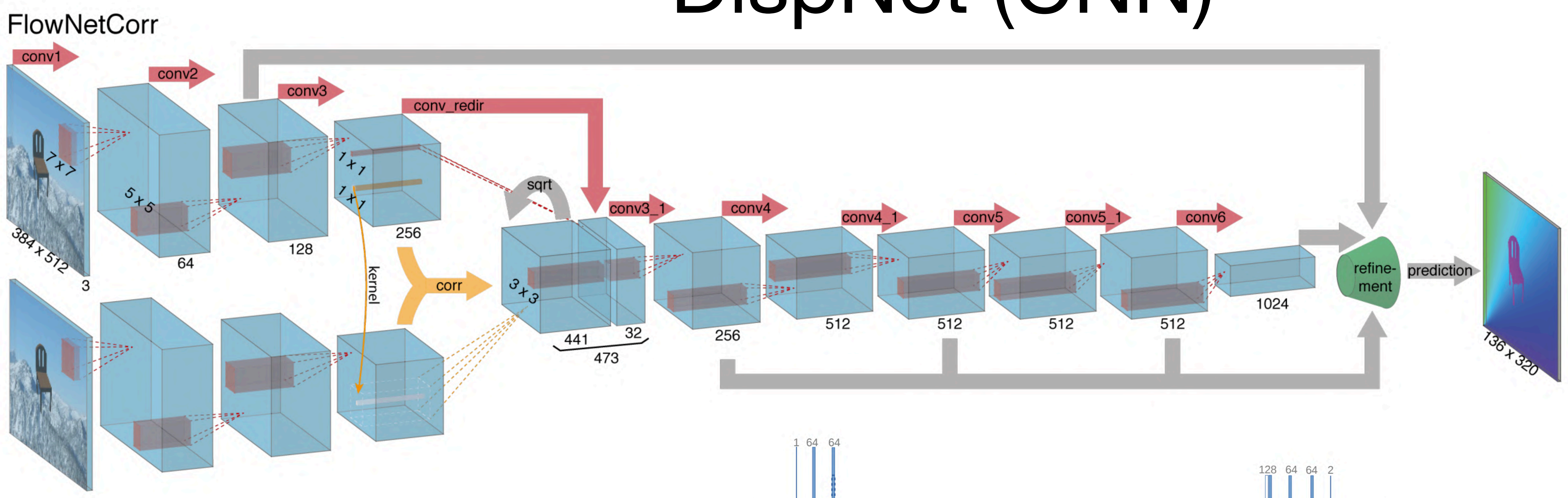
Depth Estimation in Two-View Geometry



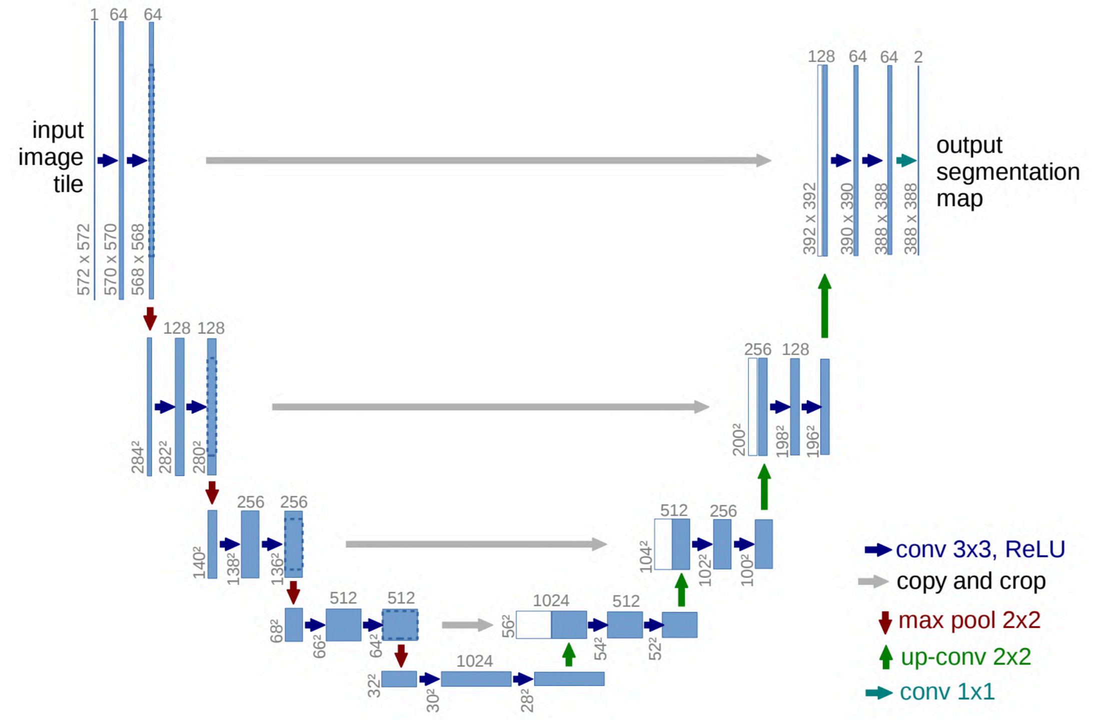
Learning-Based Approaches to Depth Estimation

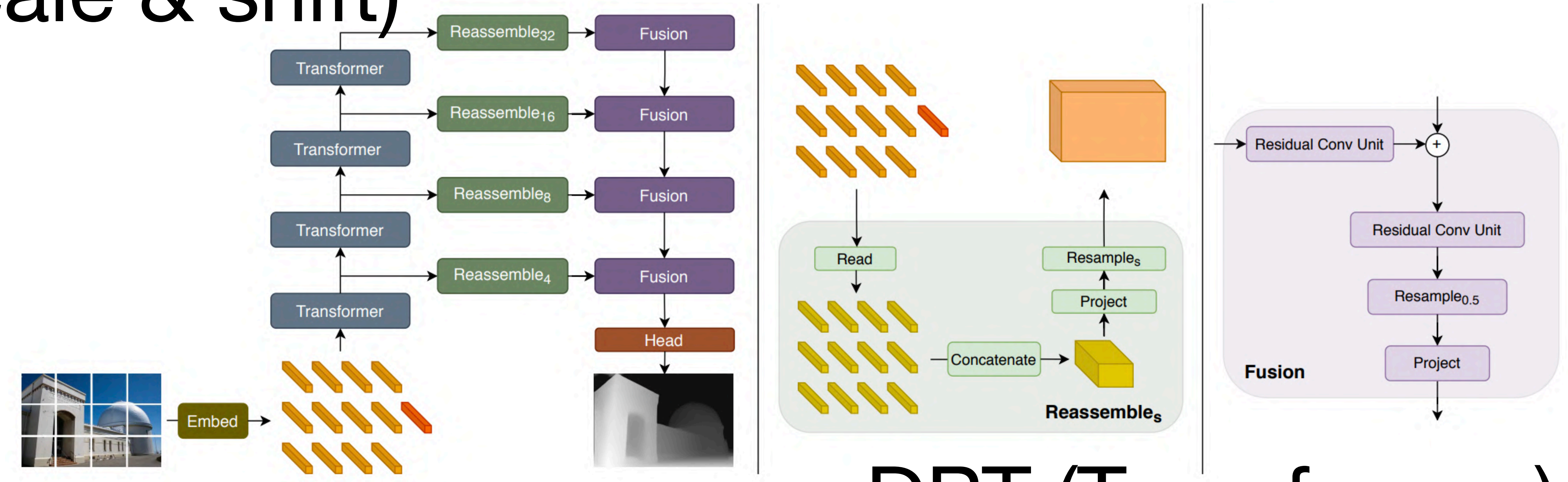
- Stereo-depth:
  - Rectification
  - Two-view architectures
- Mono-depth
  - Estimate relative depth (unknown scale & shift)
  - Dense prediction architectures

DispNet (CNN)



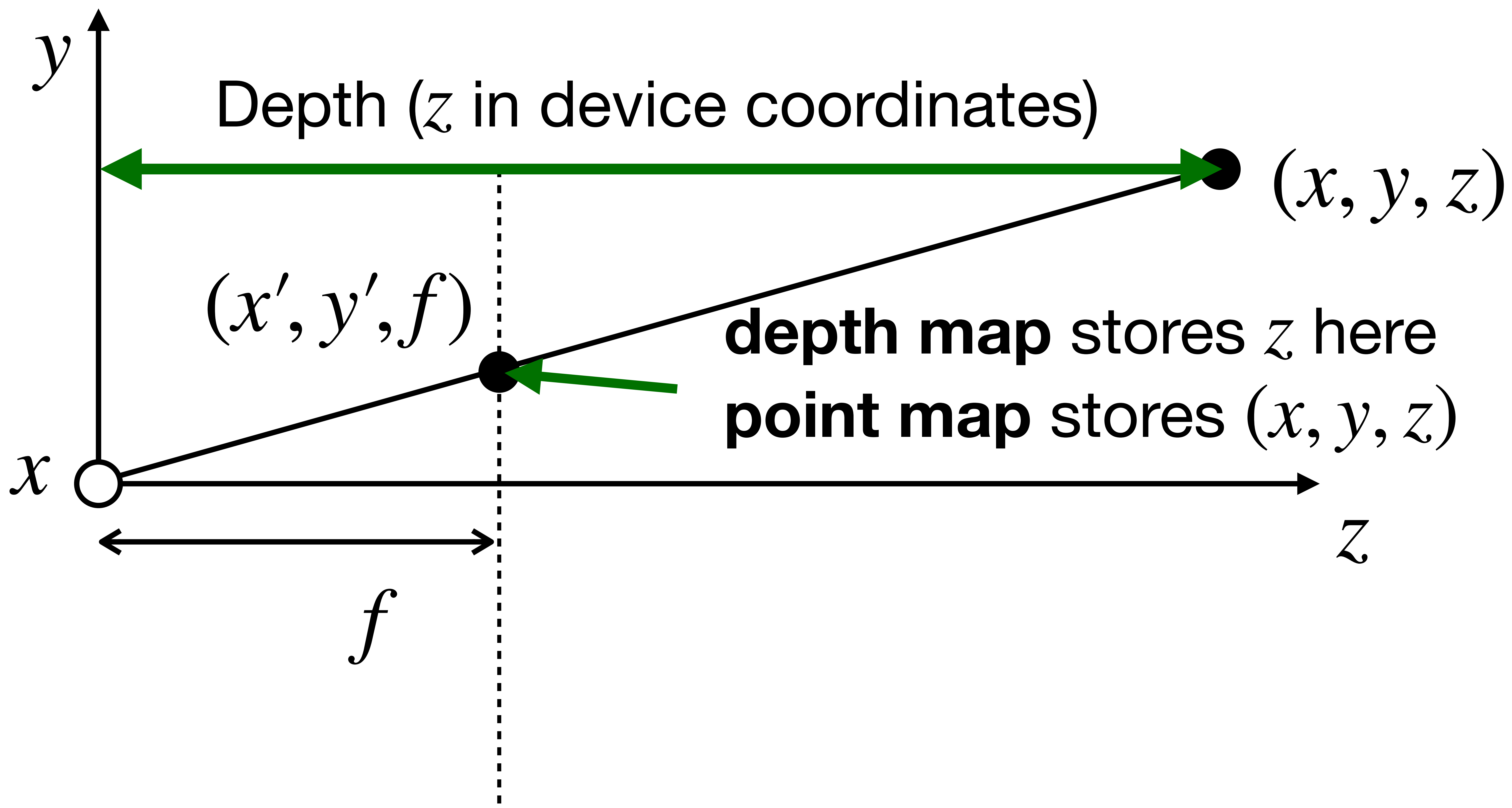
UNet (CNN)





DPT (Transformer)

Things We Tried to Predict



Depth Map

Point Map

https://github.com/isl-org/ZoeDepth
https://vgg-t.github.io/

Dust3r

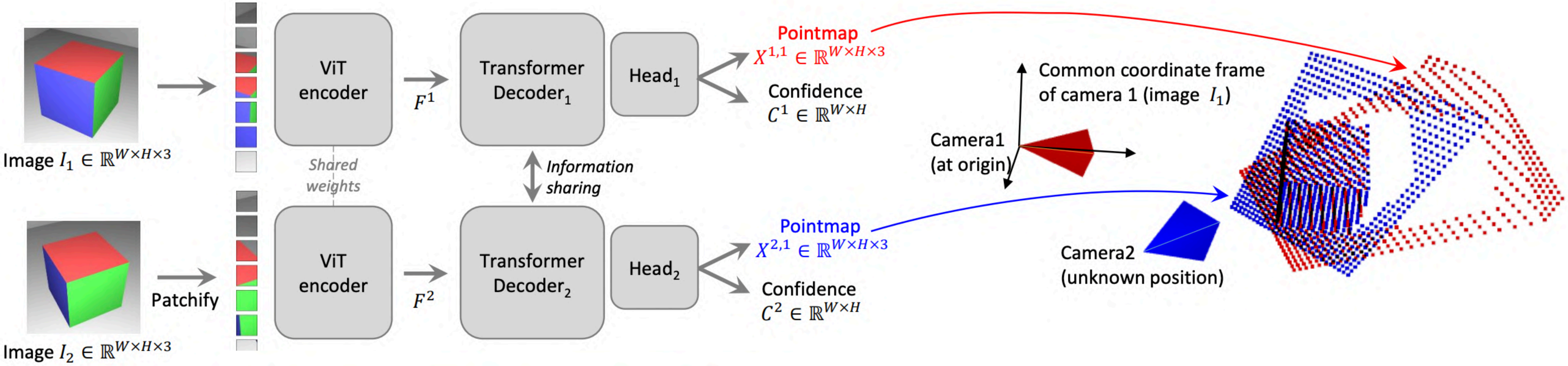
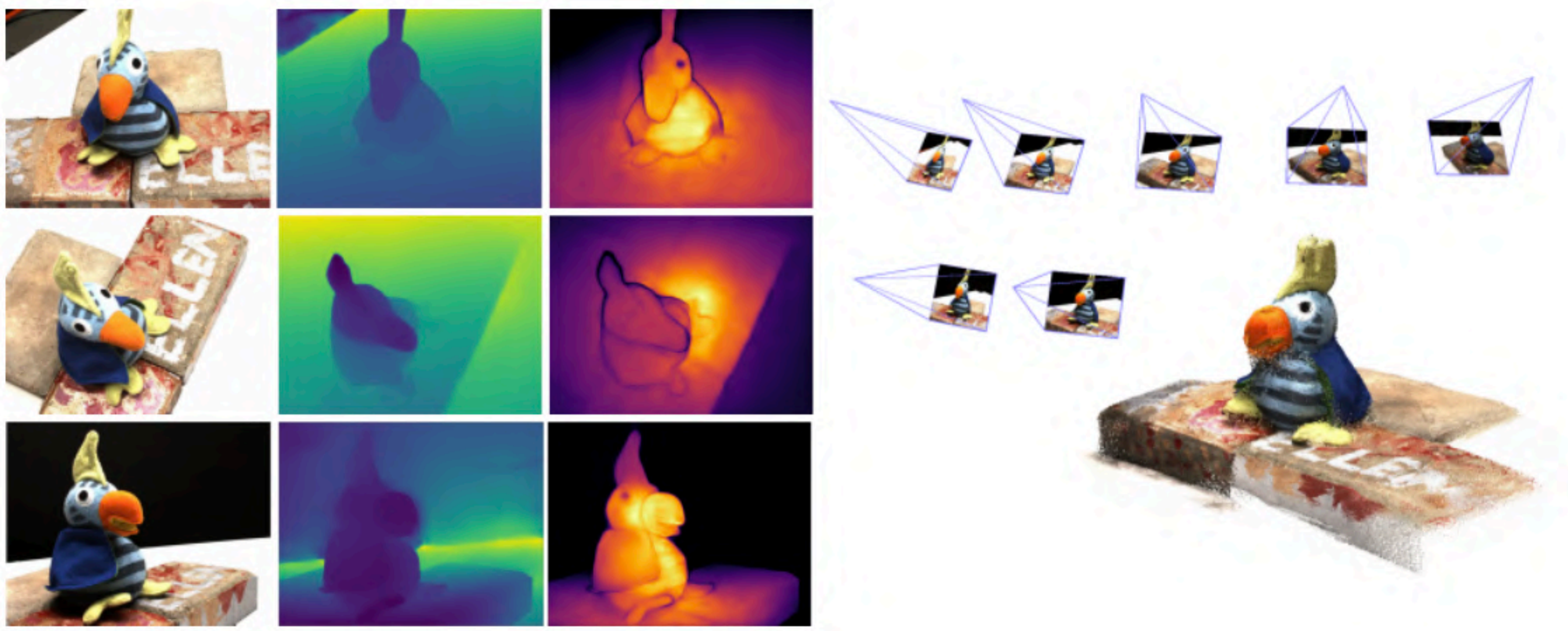


Image $I_1 \in \mathbb{R}^{W \times H \times 3}$
ViT encoder
Shared weights
F^1
Transformer Decoder$_1$
Head$_1$
Information sharing
Pointmap
$X^{1,1} \in \mathbb{R}^{W \times H \times 3}$
Confidence
$C^1 \in \mathbb{R}^{W \times H}$
Image $I_2 \in \mathbb{R}^{W \times H \times 3}$
Patchify
ViT encoder
F^2
Transformer Decoder$_2$
Head$_2$
Pointmap
$X^{2,1} \in \mathbb{R}^{W \times H \times 3}$
Confidence
$C^2 \in \mathbb{R}^{W \times H}$
Common coordinate frame of camera 1 (image I_1)
Camera1 (at origin)
Camera2 (unknown position)



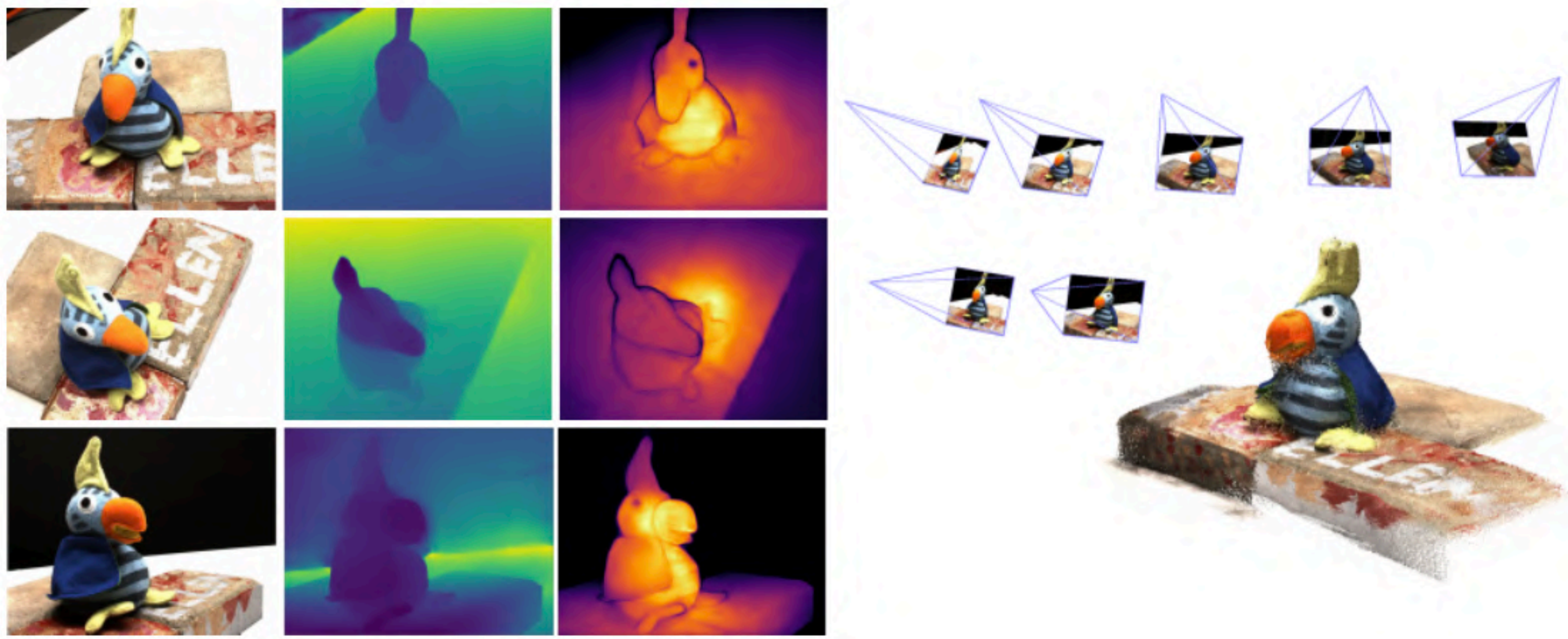
Today's Lecture: Where Do We Go Next?

- Is point map prediction enough?
- Shall we limit ourselves to two-view models?
- Elephant in the room: we live in dynamic world

- Today we will discuss further developments in foundational 3D CV models
  - *Note: This overview is selective and focuses on key works discussed in detail.*

Delving Deeper into Dust3R

- Single objective: estimate point maps
- Point map estimates are imperfect
    - DTH dataset evaluates multi-view reconstruction
    - Measures distance between estimated & ground truth point clouds



	Methods	GT cams	Acc.↓	Comp.↓	Overall↓
(a)	Camp [12]	✓	0.835	0.554	0.695
	Furu [33]	✓	0.613	0.941	0.777
	Tola [134]	✓	0.342	1.190	0.766
	Gipuma [34]	✓	**0.283**	0.873	0.578
(b)	MVSNet [161]	✓	0.396	0.527	0.462
	CVP-MVSNet [158]	✓	0.296	0.406	0.351
	UCS-Net [18]	✓	0.338	0.349	0.344
	CER-MVS [65]	✓	0.359	0.305	0.332
	CIDER [157]	✓	0.417	0.437	0.427
	CasMVSNet [40]	✓	0.325	0.385	0.355
	PatchmatchNet [139]	✓	0.427	0.277	0.352
	GeoMVSNet [180]	✓	0.331	**0.259**	**0.295**
	DUSt3R 512	×	2.677	0.805	1.741

Delving Deeper into Dust3R

- There are two ways to estimate camera parameters
- Old way:
  - Compute pixel matches based on points maps
  - Use two-view geometry tools to estimate cameras based on matches
- New way: estimate projection matrices based on point-maps
- Median translation/rotation estimation error in multi-view camera estimation:

Methods	GT	7Scenes (Indoor) [114]							Cambridge (Outdoor) [49]				
	Focals	Chess	Fire	Heads	Office	Pumpkin	Kitchen	Stairs	S. Facade	O. Hospital	K. College	St.Mary's	G. Court
DUSt3R 512 from 2D-matching	✓	3/0.97	3/0.95	2/1.37	**3**/1.01	**4**/1.14	4/1.34	11/2.84	6/0.26	17/0.33	**11/0.20**	**7/0.24**	38/0.16
DUSt3R 512 from scaled rel-pose	×	5/1.08	5/1.18	4/1.33	6/1.05	7/1.25	6/1.37	26/3.56	64/0.97	151/0.88	102/0.88	79/1.46	245/1.08

Mast3R

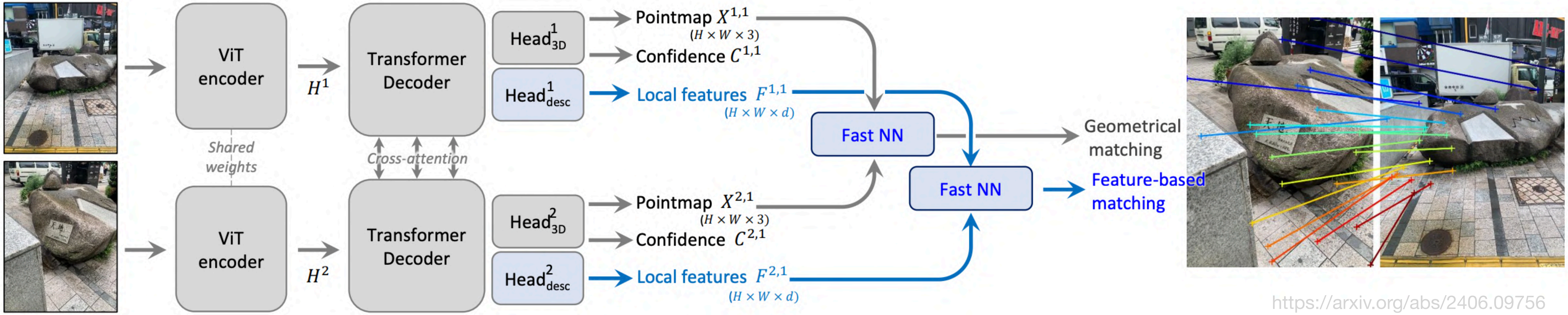
Train a Separate Head for Matching

- Dust3R computed matches based on adjacency 3D space
- Mast3R computes matches based on feature-matching

$$\mathcal{L}_{\text{match}} = - \sum_{(i,j)\in\hat{\mathcal{M}}} \log \frac{s_\tau(i,j)}{\sum_{k\in\mathcal{P}^1} s_\tau(k,j)} + \log \frac{s_\tau(i,j)}{\sum_{k\in\mathcal{P}^2} s_\tau(i,k)}$$

(InfoNCE)

$$\text{with } s_\tau(i,j) = \exp\left[-\tau D_i^{1\top} D_j^2\right].$$

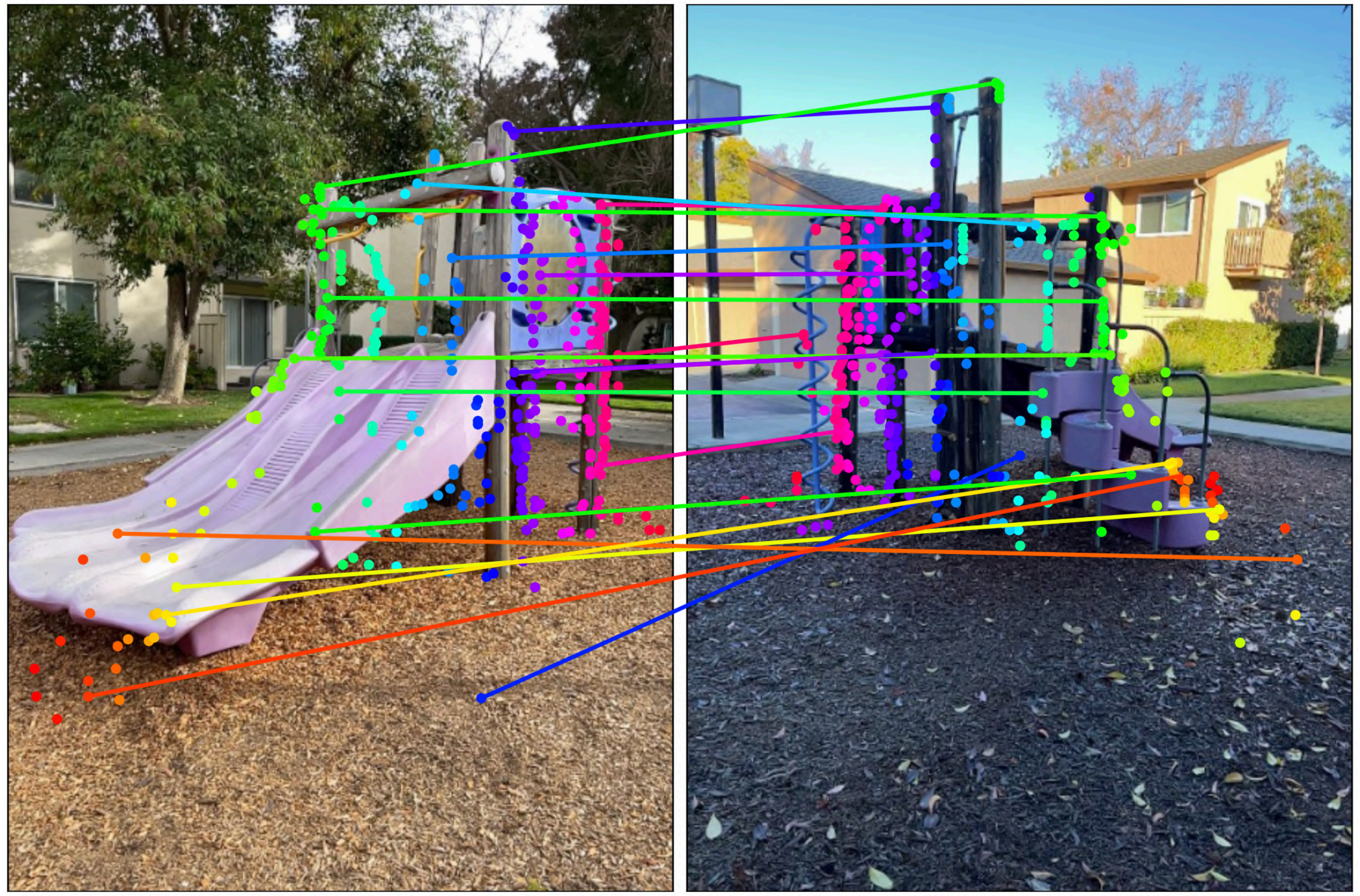


https://arxiv.org/abs/2406.09756

Mast3R

Map-Free Relocalization Challenge

- Estimate relative rotation & **metric** translation for a given pair of images



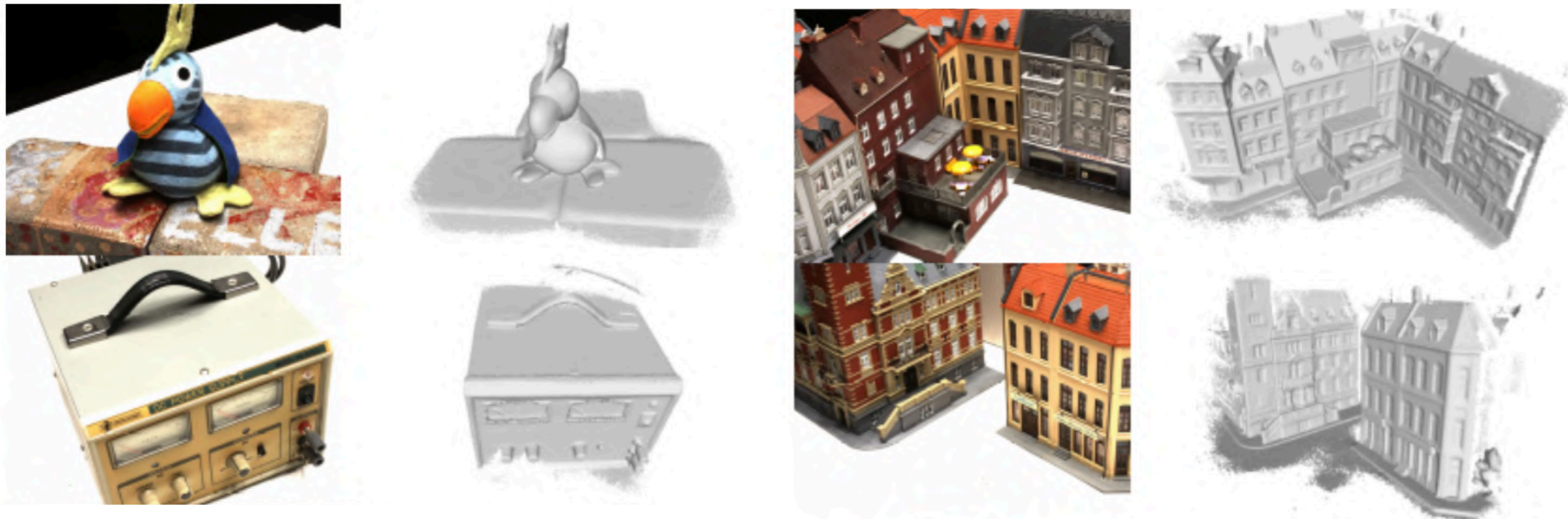
Method	AUC (VCRE < 45px)	AUC (VCRE < 90px)	Precision (VCRE < 45px)	Precision (VCRE < 90px)	Median Reproj. Error (px)
MASt3R (Ess.Mat + D.Scale)	0.817	0.933	63.0%	79.3%	48.8
interp_metric3d_loftr_3d2d	0.681	0.796	39.9%	51.4%	125.6
Map-Free Visual Relocalization Enhanced by Instance Knowledge and Depth Knowledge	0.656	0.849	45.3%	65.0%	80.0
RoMa w/ MicKey depth maps	0.604	0.734	37.8%	52.4%	111.9
SuperGlue w/ MicKey depth maps	0.556	0.711	29.8%	43.0%	139.9
LoFTR w/ MicKey depth maps	0.550	0.715	27.2%	40.7%	155.0
Mickey Variant GT_Depth	0.548	0.720	28.0%	45.5%	142.0

https://nianticlabs.github.io/map-free-workshop/2024/

Mast3R

Multi-View Reconstruction Improvements

- Dust3R: join point maps to get point cloud
- Mast3R improves accuracy:
    - Join matching points
    - Triangulate points to get a mesh



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	Furu [31]	0.613	0.941	0.777
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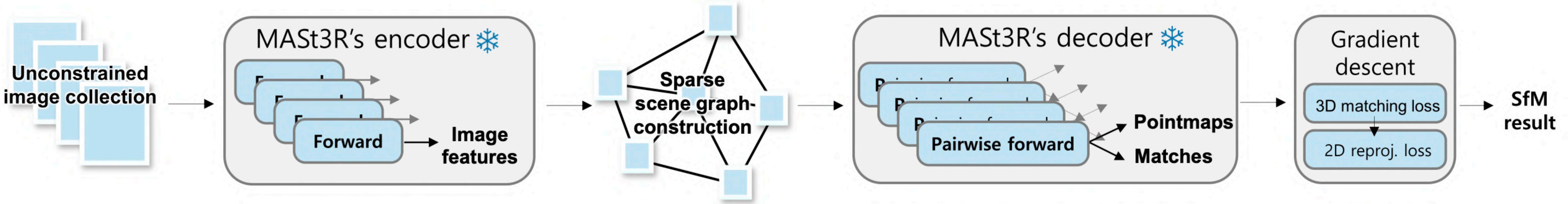
(d)	MVSNet [110]	0.396	0.527	0.462
	CVP-MVSNet [109]	0.296	0.406	0.351
	UCS-Net [17]	0.338	0.349	0.344
	CER-MVS [55]	0.359	0.305	0.332
	CIDER [107]	0.417	0.437	0.427
	PatchmatchNet [99]	0.427	0.277	0.352
	GeoMVSNet [119]	0.331	**0.259**	**0.295**
(e)	DUSt3R [102]	2.677	0.805	1.741
	MASt3R	0.403	0.344	0.374

Mast3R-SfM

From Two-View to Multiple View

- Naive approach: run Mast3R on all image pairs to get cameras & point cloud
- Incremental structure from motion builds a scene graph and adds images bit by bit
- Mast3R-SfM proposes an approach to build a scene graph based on Mast3R



https://arxiv.org/abs/2409.19152

Mast3R-SfM

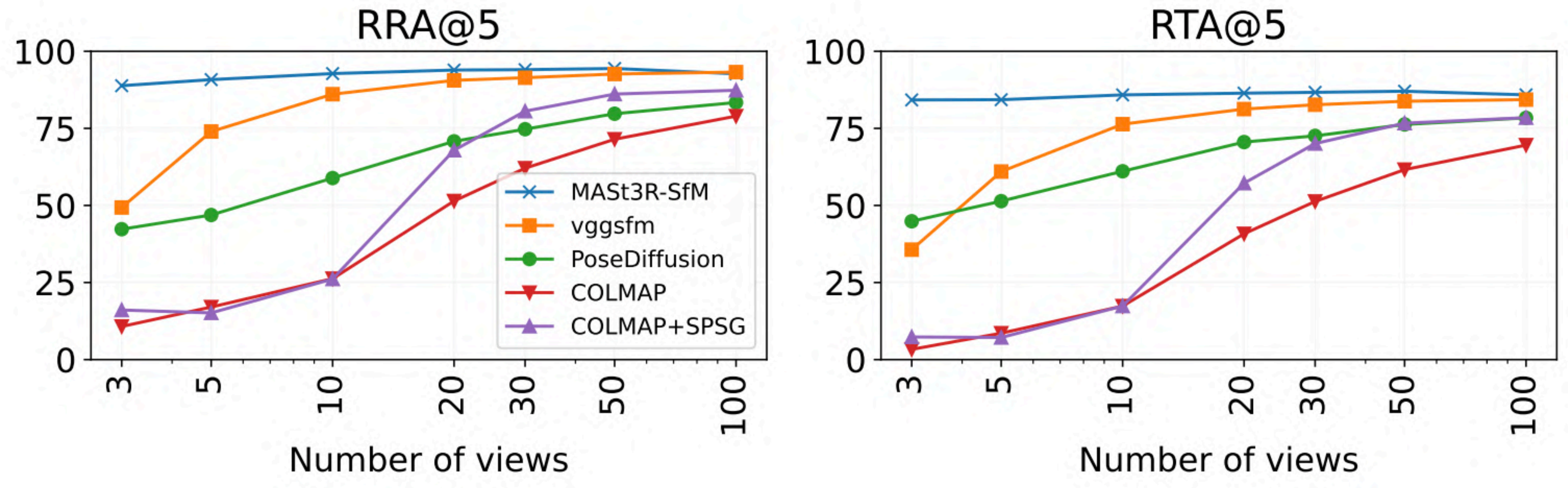
Evaluations

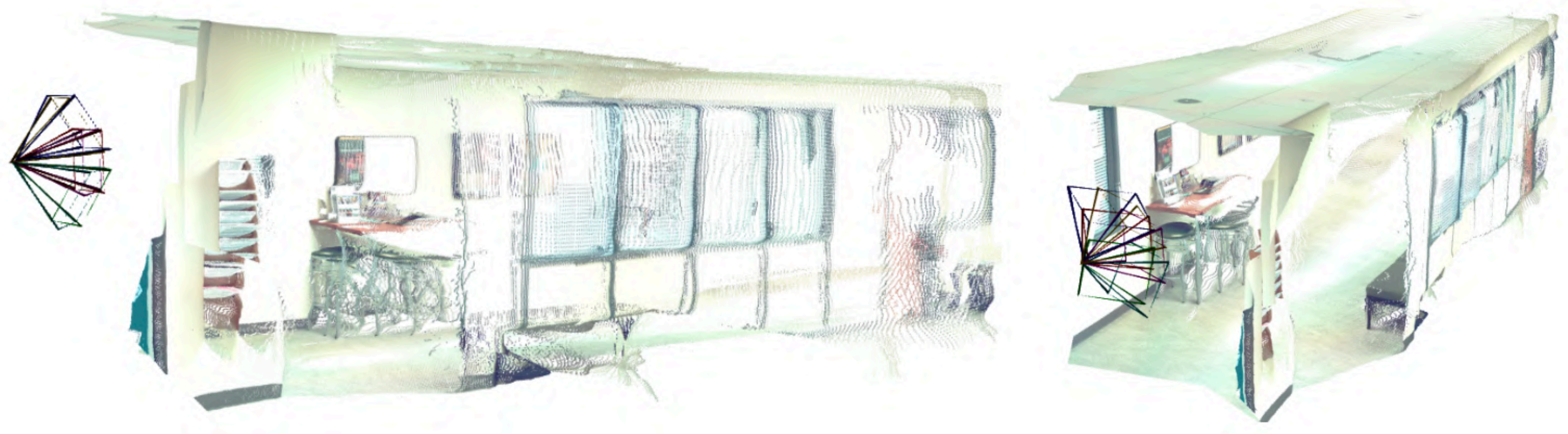
Metric glossary:

RRA@5 - relative rotation accuracy at 5°

RTA@5 - relative translation accuracy at 5°

ATE - absolute trajectory error





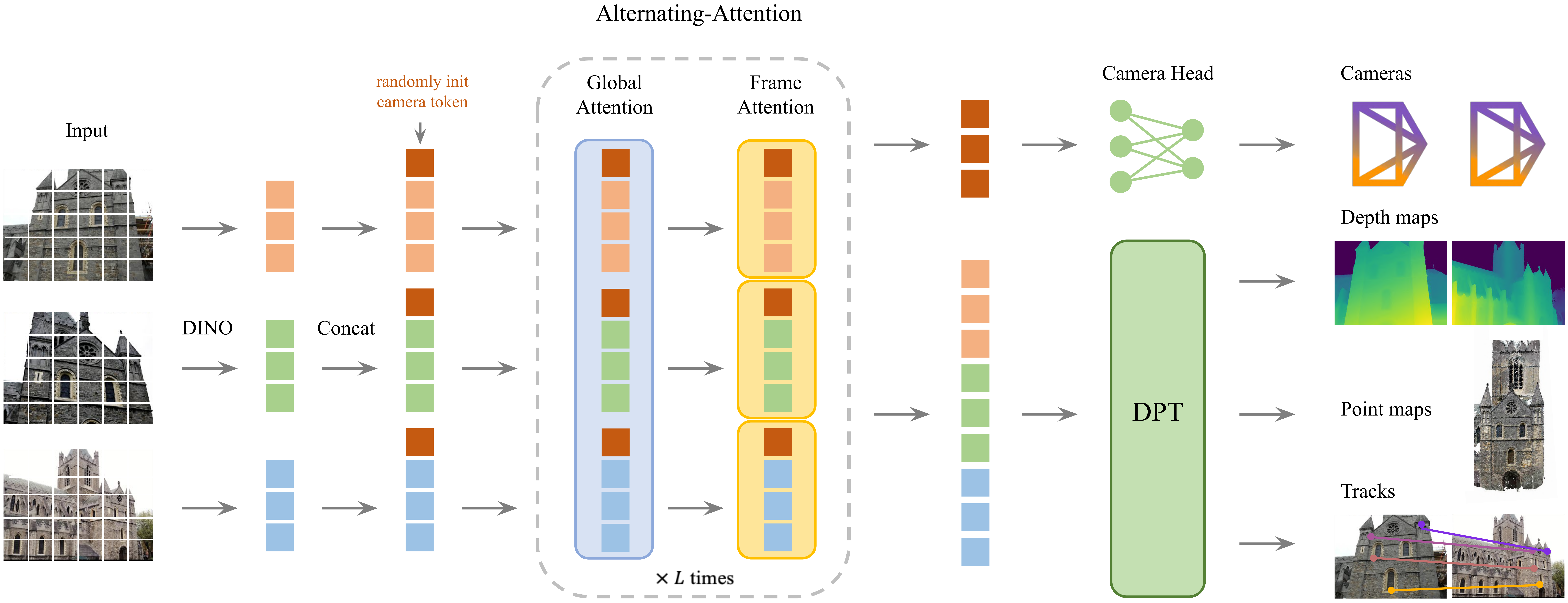
Method	25 views		50 views		100 views		200 views		full	
	ATE↓	Reg.↑	ATE↓	Reg.↑	ATE↓	Reg.↑	ATE↓	Reg.↑	ATE↓	Reg.↑
COLMAP [46]	0.03840	44.4	0.02920	60.5	0.02640	85.7	0.01880	97.0	-	-
ACE-Zero [9]	0.11160	100.0	0.07130	100.0	0.03980	100.0	0.01870	100.0	0.01520	100.0
FlowMap [50]	0.10700	100.0	0.07310	100.0	0.04460	100.0	0.02420	100.0	N/A	66.7
VGGSfM [62]	0.05800	96.2	0.03460	98.7	0.02900	98.5	N/A	47.6	N/A	0.0
DF-SfM [20]	0.08110	99.4	0.04120	100.0	0.02710	99.9	N/A	33.3	N/A	76.2
MASt3R-SfM	0.03360	100.0	0.02610	100.0	0.01680	100.0	0.01300	100.0	0.01060	100.0

Scenes	COLMAP [46]		ACE-Zero [9]		FlowMap [50]		VGGSfM [62]		DF-SfM [20]		**MASt3R-SfM**	
	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5
courtyard	56.3	60.0	4.0	1.9	7.5	3.6	50.5	51.2	80.7	74.8	89.8	64.4
delivery area	34.0	28.1	27.4	1.9	29.4	23.8	22.0	19.6	82.5	82.0	83.1	81.8
electro	53.3	48.5	16.9	7.9	2.5	1.2	79.9	58.6	82.8	81.2	100.0	95.5
facade	92.2	90.0	74.5	64.1	15.7	16.8	57.5	48.7	80.9	82.6	74.3	75.3
kicker	87.3	86.2	26.2	16.8	1.5	1.5	100.0	97.8	93.5	91.0	100.0	100.0
meadow	0.9	0.9	3.8	0.9	3.8	2.9	100.0	96.2	56.2	58.1	58.1	58.1
office	36.9	32.3	0.9	0.0	0.9	1.5	64.9	42.1	71.1	54.5	100.0	98.5
pipes	30.8	28.6	9.9	1.1	6.6	12.1	100.0	97.8	72.5	61.5	100.0	100.0
playground	17.2	18.1	3.8	2.6	2.6	2.8	37.3	40.8	70.5	70.1	100.0	93.6
relief	16.8	16.8	16.8	17.0	6.9	7.7	59.6	57.9	32.9	32.9	34.2	40.2
relief 2	11.8	11.8	7.3	5.6	8.4	2.8	69.9	70.3	40.9	39.1	57.4	76.1
terrace	100.0	100.0	5.5	2.0	33.2	24.1	38.7	29.6	100.0	99.6	100.0	100.0
terrains	100.0	99.5	15.8	4.5	12.3	13.8	70.4	54.9	100.0	91.9	58.2	52.5
Average	49.0	47.8	16.4	9.7	10.1	8.8	65.4	58.9	74.2	70.7	81.2	79.7

VGGT: Visual Geometry Grounded Transformer

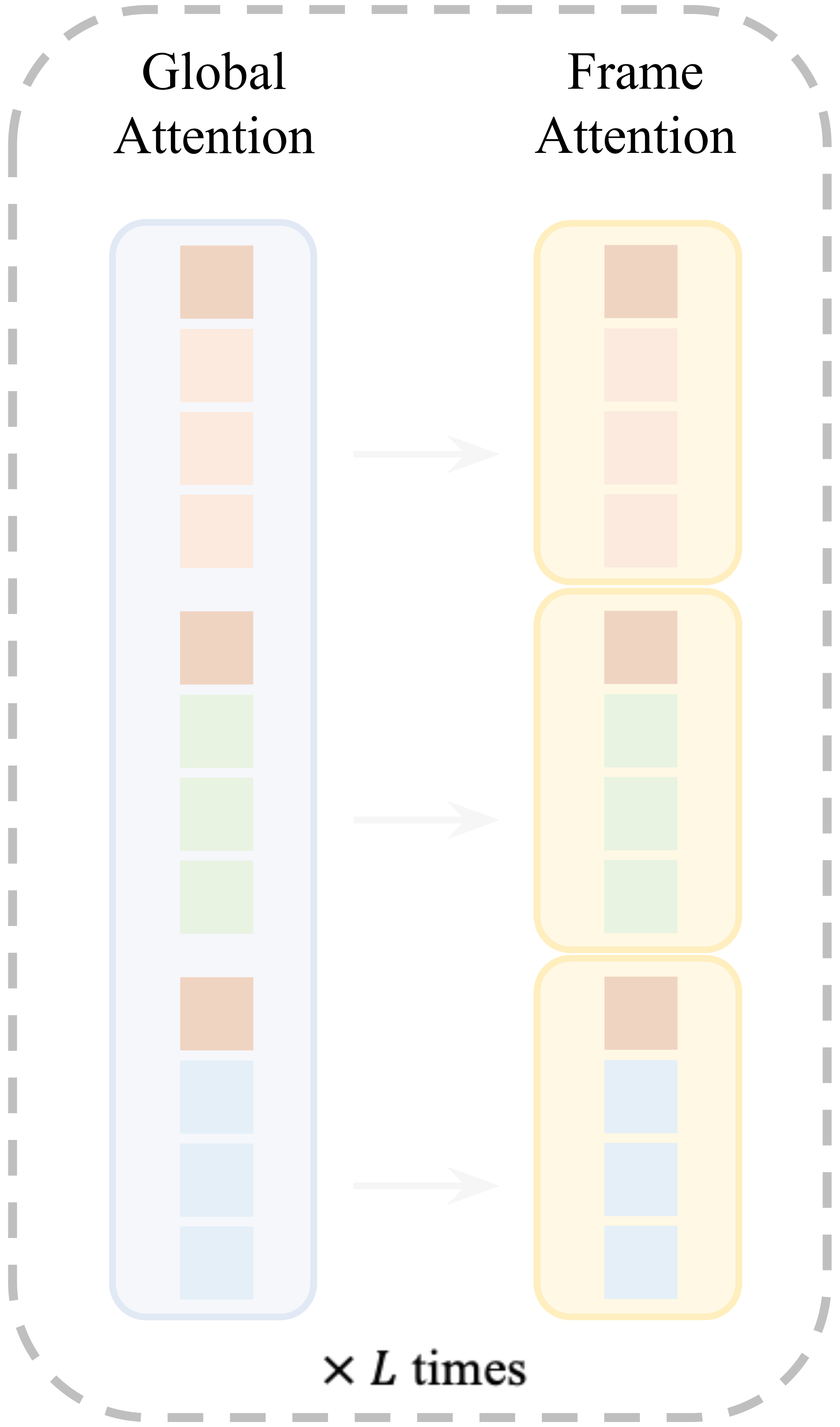
Following slides are taken from https://vgg-t.github.io/

VGG Transformer

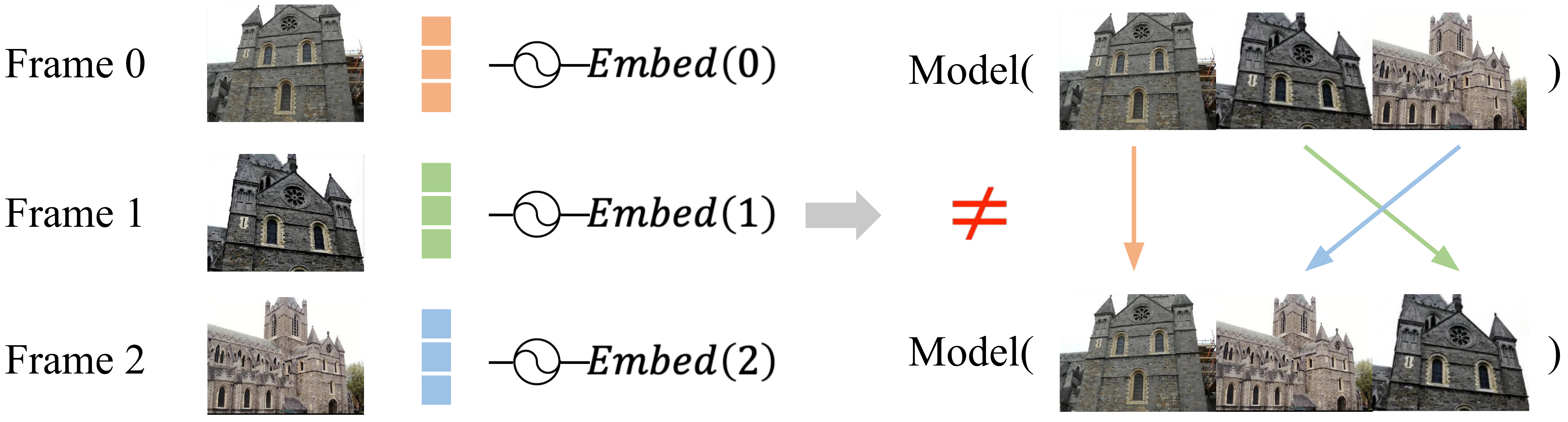


Why Alternating-Attention?

- Global Attention
  - Ensures scene-level coherence
- Frame-wise Attention
  - Eliminates **frame index embedding**
    - For permutation equivariance
    - For flexible input length



Why Alternating-Attention?



Not permutation equivariant

Why Alternating-Attention?

Frame 0  $-\!\!\odot\!\!-Embed(0)$

Frame 1  $-\!\!\odot\!\!-Embed(1)$

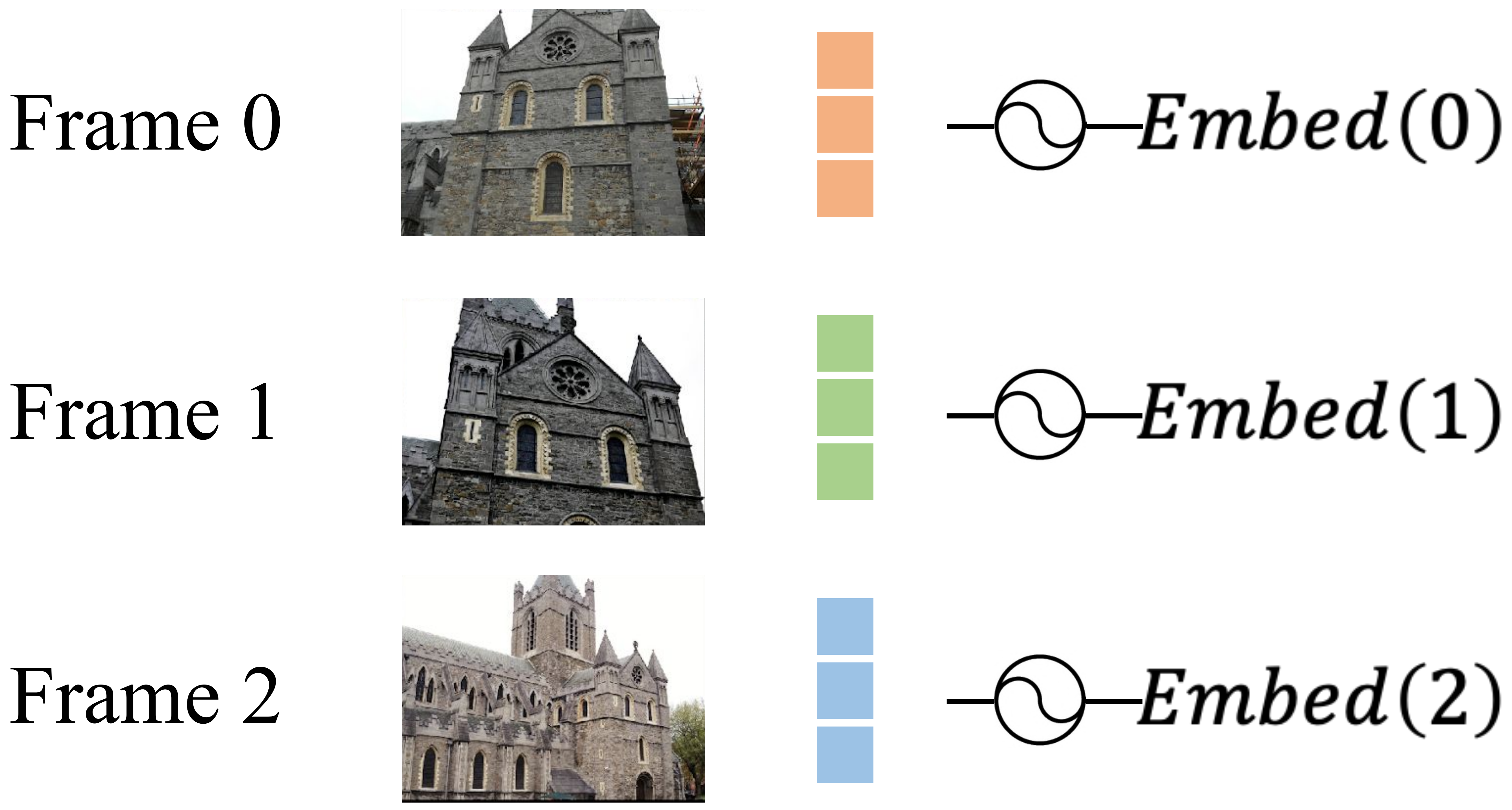
Frame 2  $-\!\!\odot\!\!-Embed(2)$

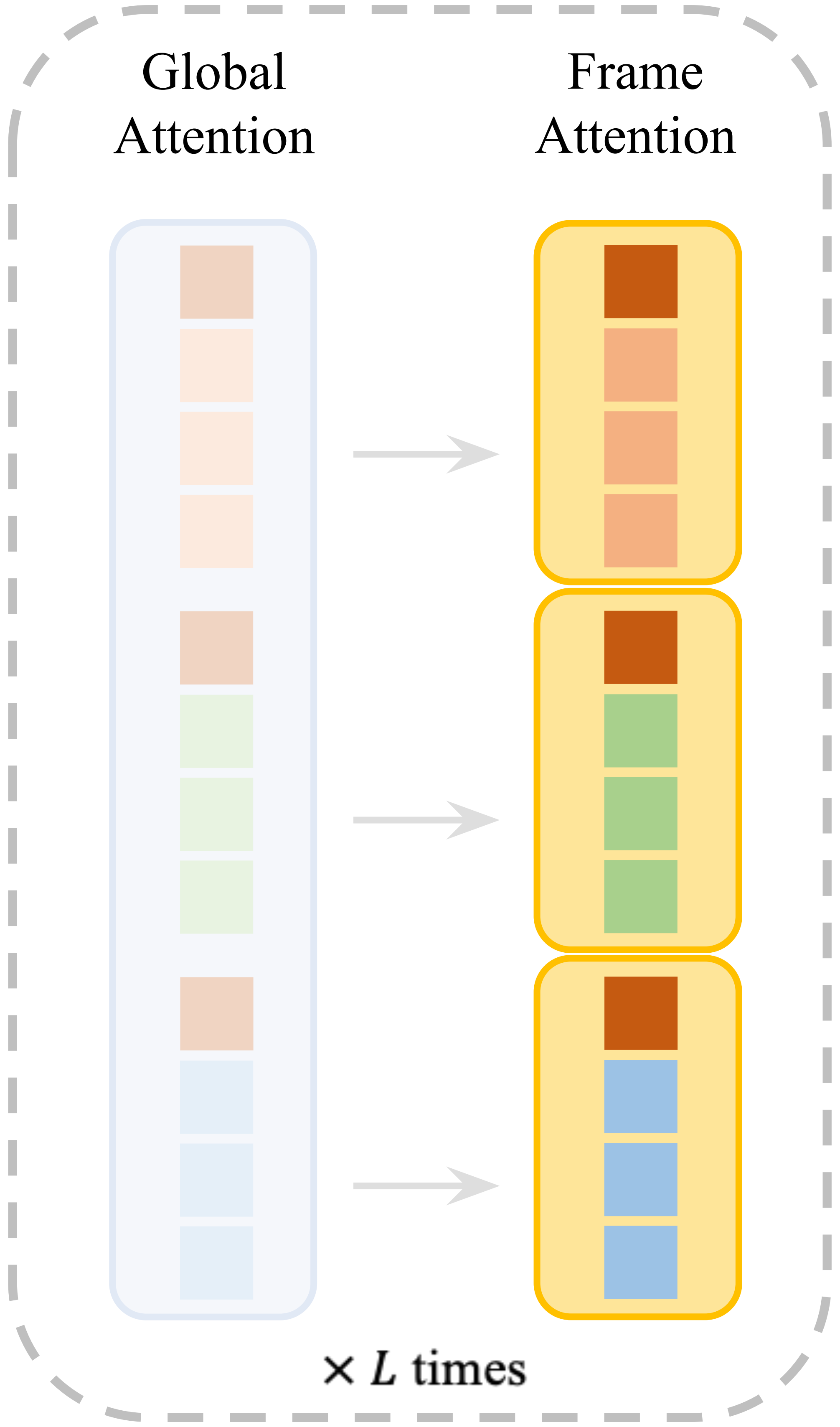
⋮

Frame 842

But model never sees $Embed(842)$ during training

Why Alternating-Attention?



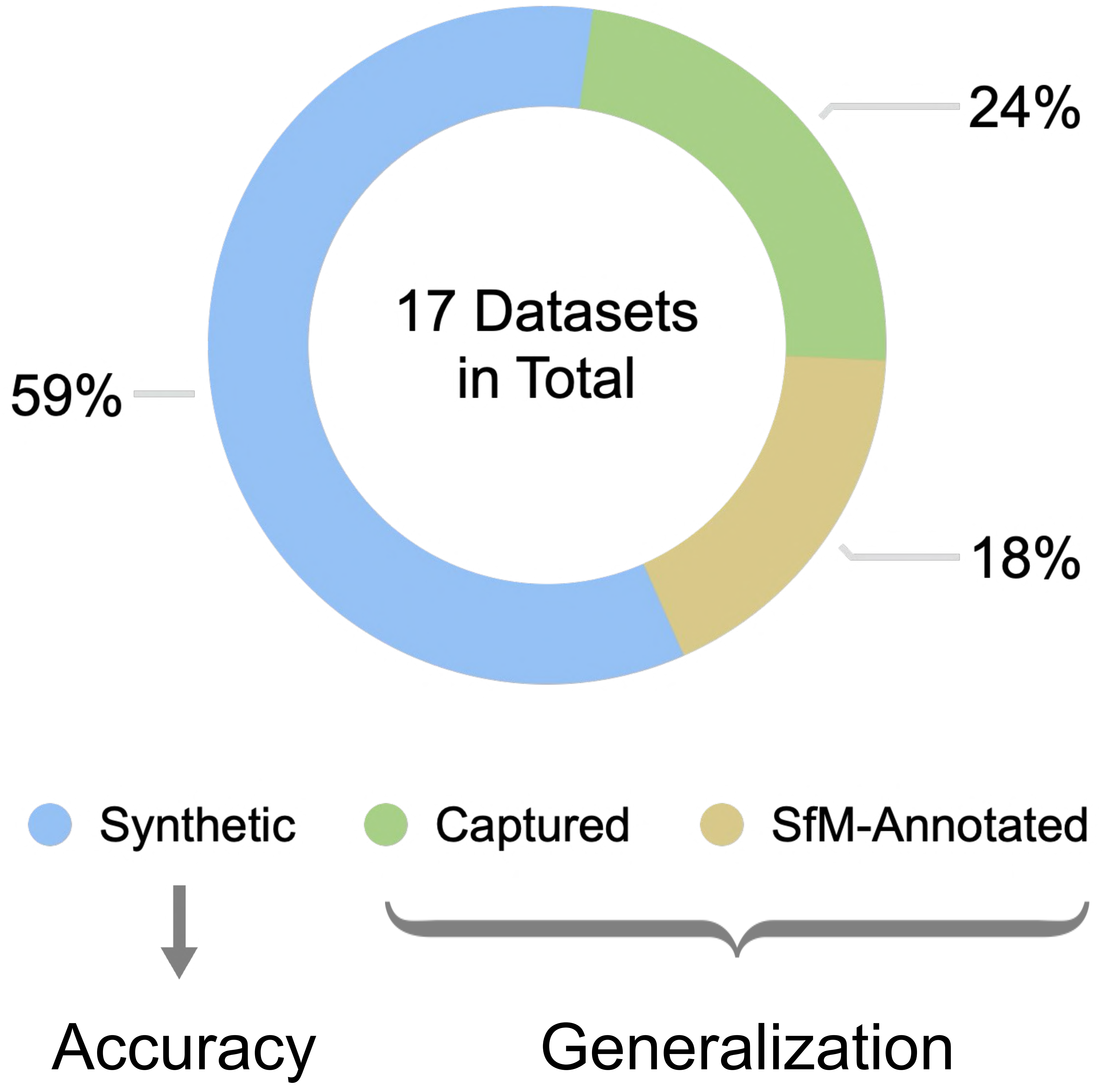


Replaces frame index embedding by Frame-wise Attention

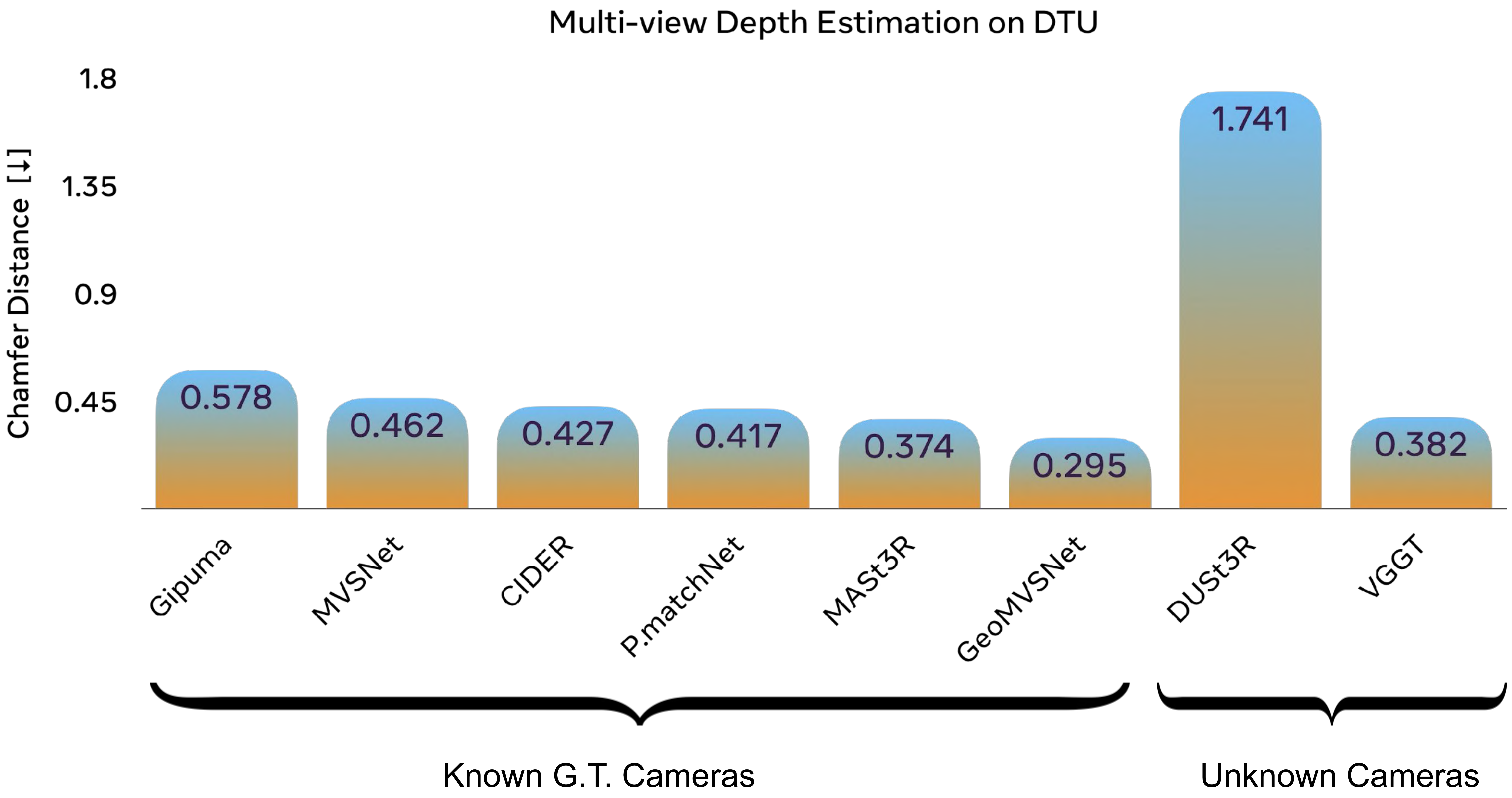
Training and Data

💡 Training:
2 to 24 frames

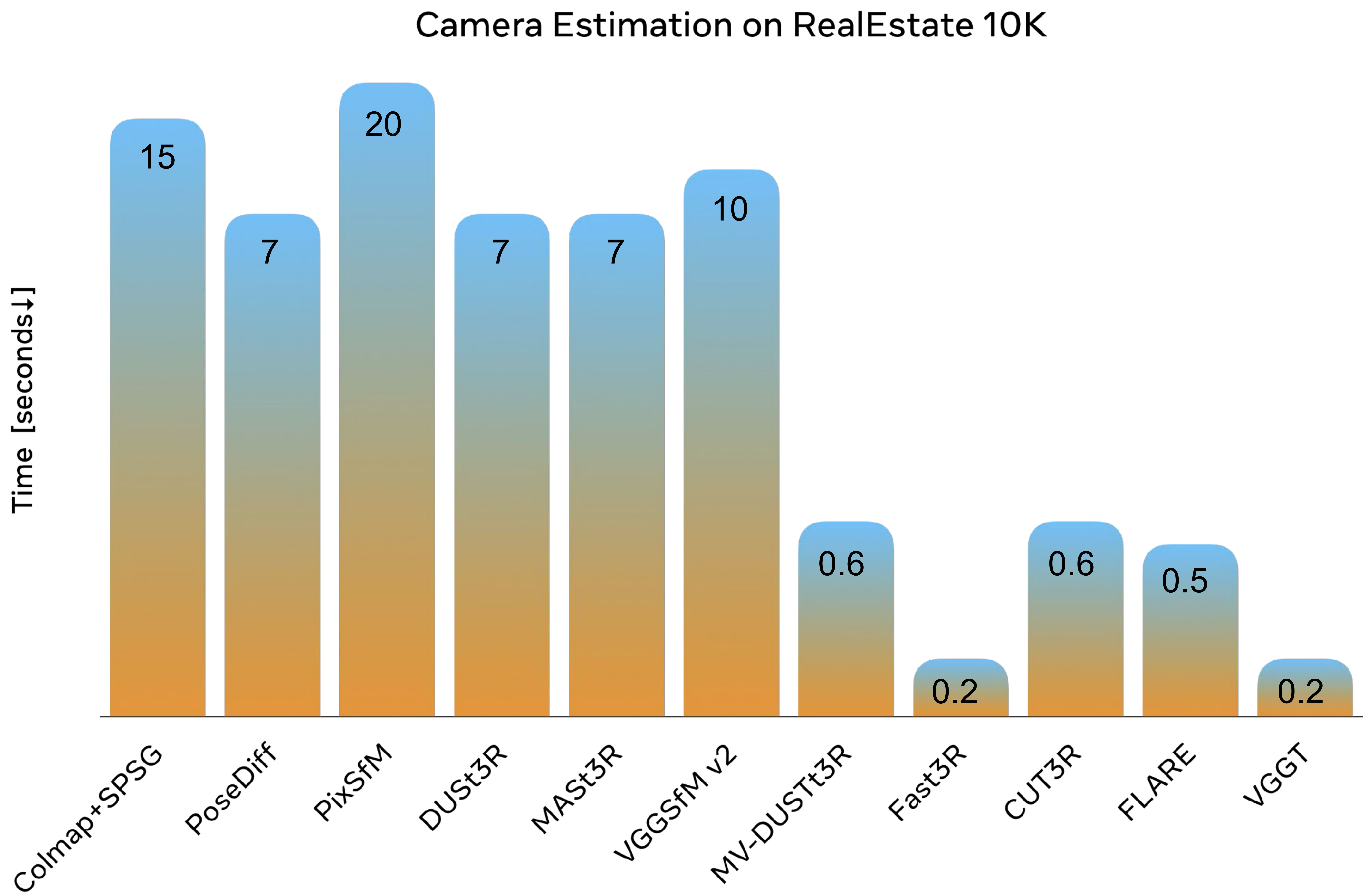
🔍 Inference:
1 to 300+ frames



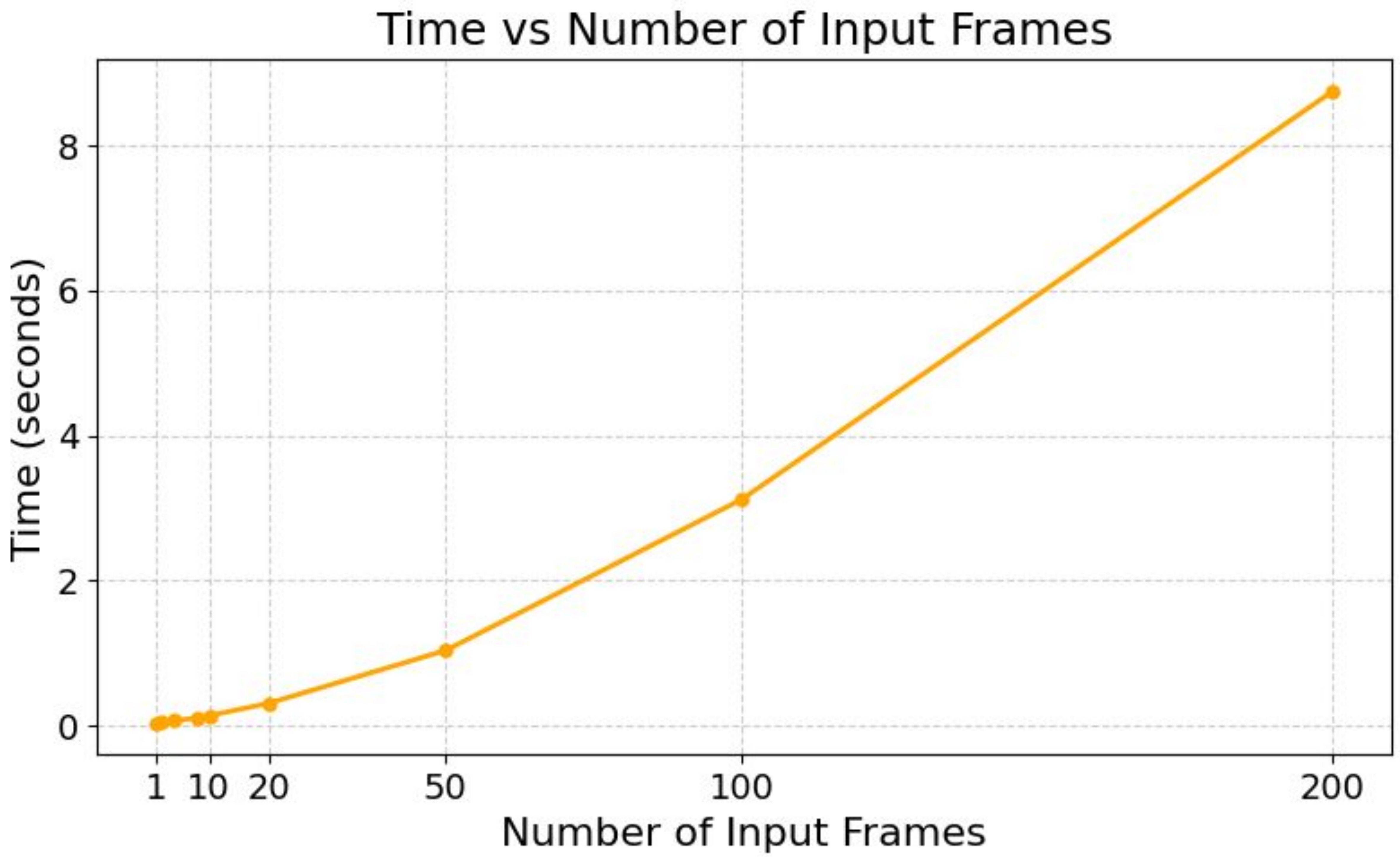
VGGT Is Accurate

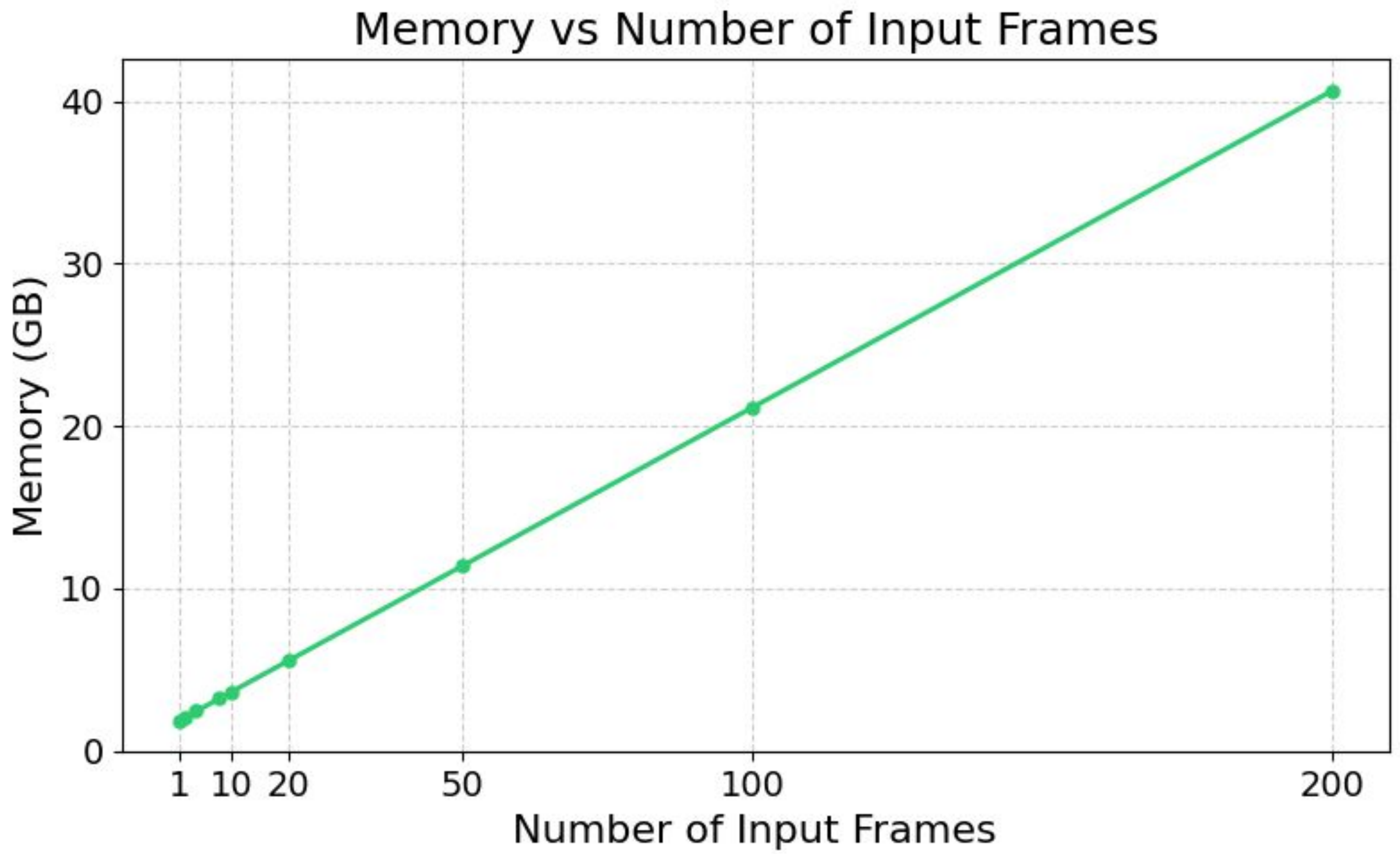


VGGT Is Fast


Camera Estimation on RealEstate 10K
Time [seconds↓]
15
7
20
7
7
10
0.6
0.2
0.6
0.5
0.2
Colmap+SPSG
PoseDiff
PixSfM
DUSt3R
MASt3R
VGGSfM v2
MV-DUSTt3R
Fast3R
CUT3R
FLARE
VGGT

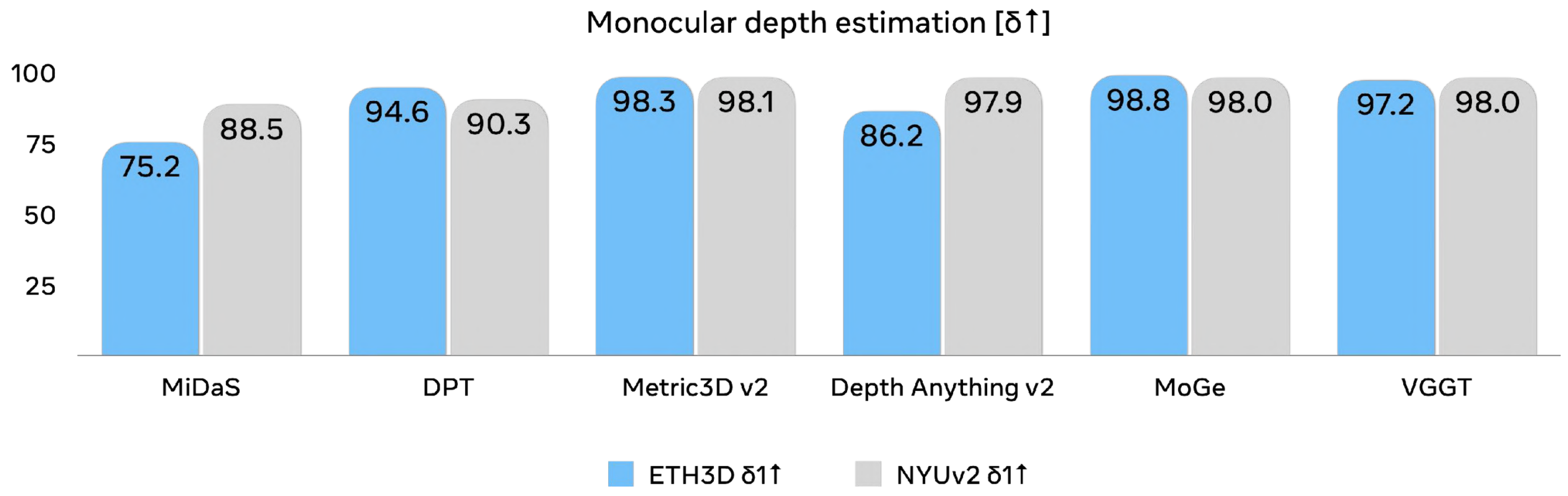
Runtime and Memory





- Memory usage scales roughly linearly with input frames
- The time usage is around $O(N^{1.5})$

Zero-shot Monocular Depth Estimation



As good as SoTA experts – but VGGT was never trained for monocular

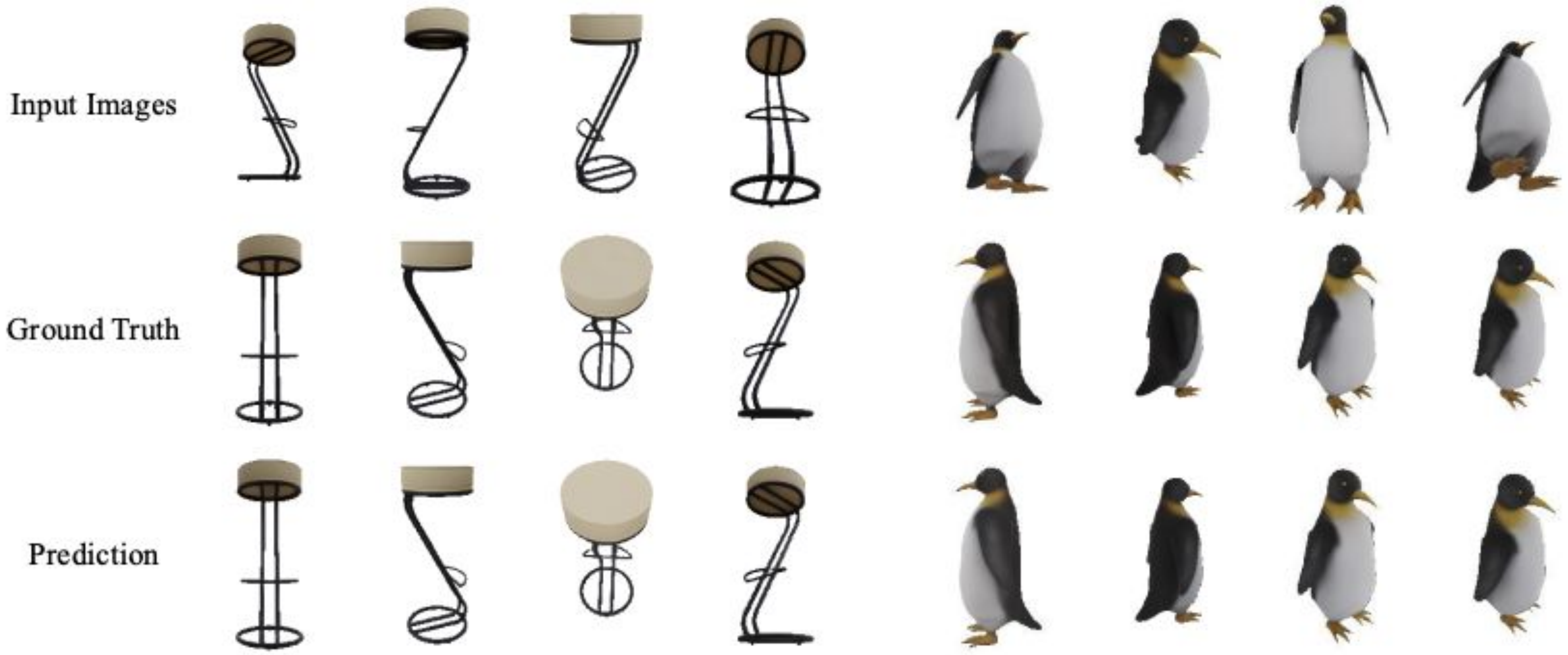
VGGT Helps Downstream Tasks

Dynamic Point Tracking

VGGT's feature backbone

boosts CoTracker to SoTA

Novel View Synthesis



VGGT enables NVS without camera inputs

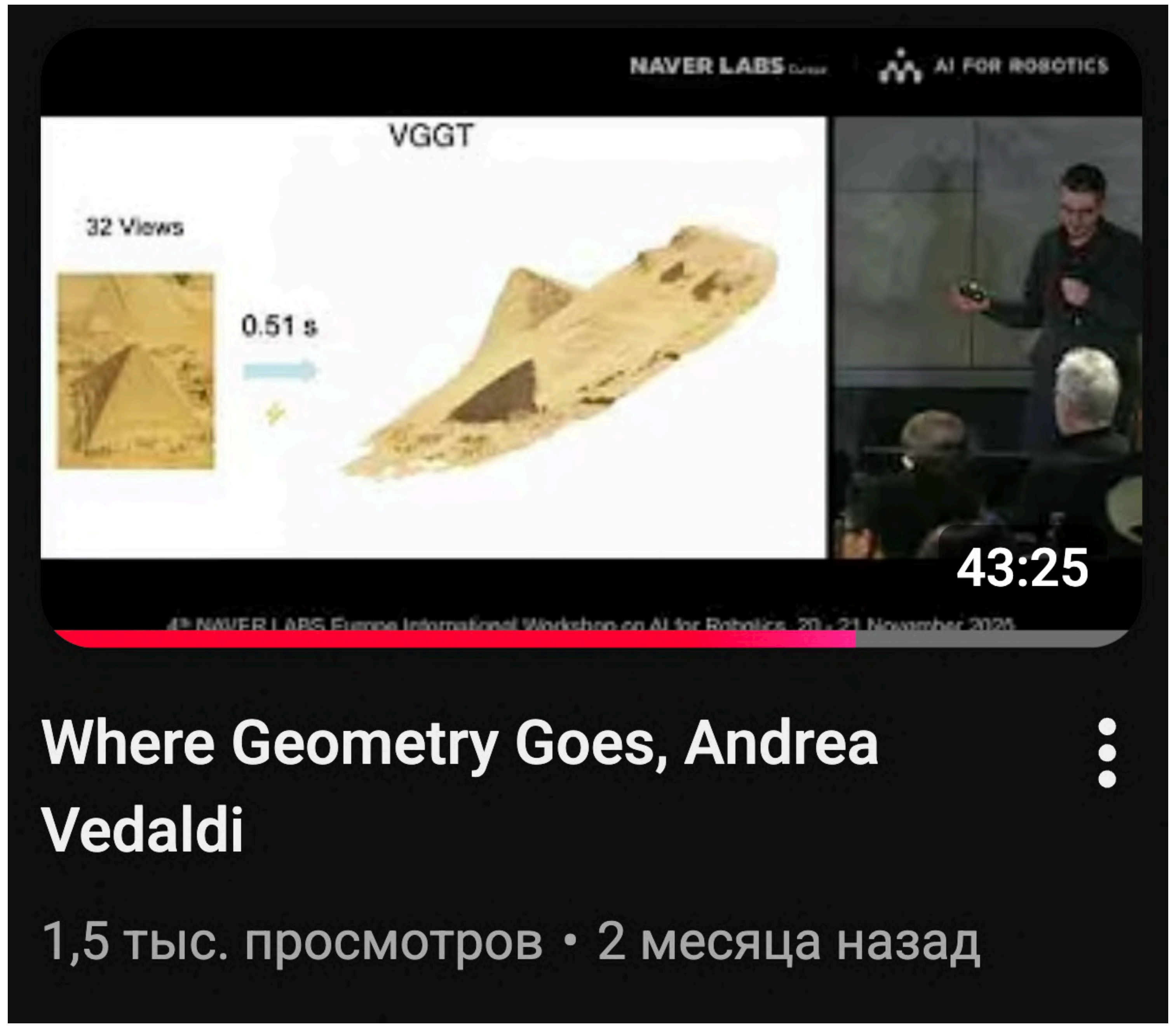
retaining comparable quality

Going Beyond Static Data

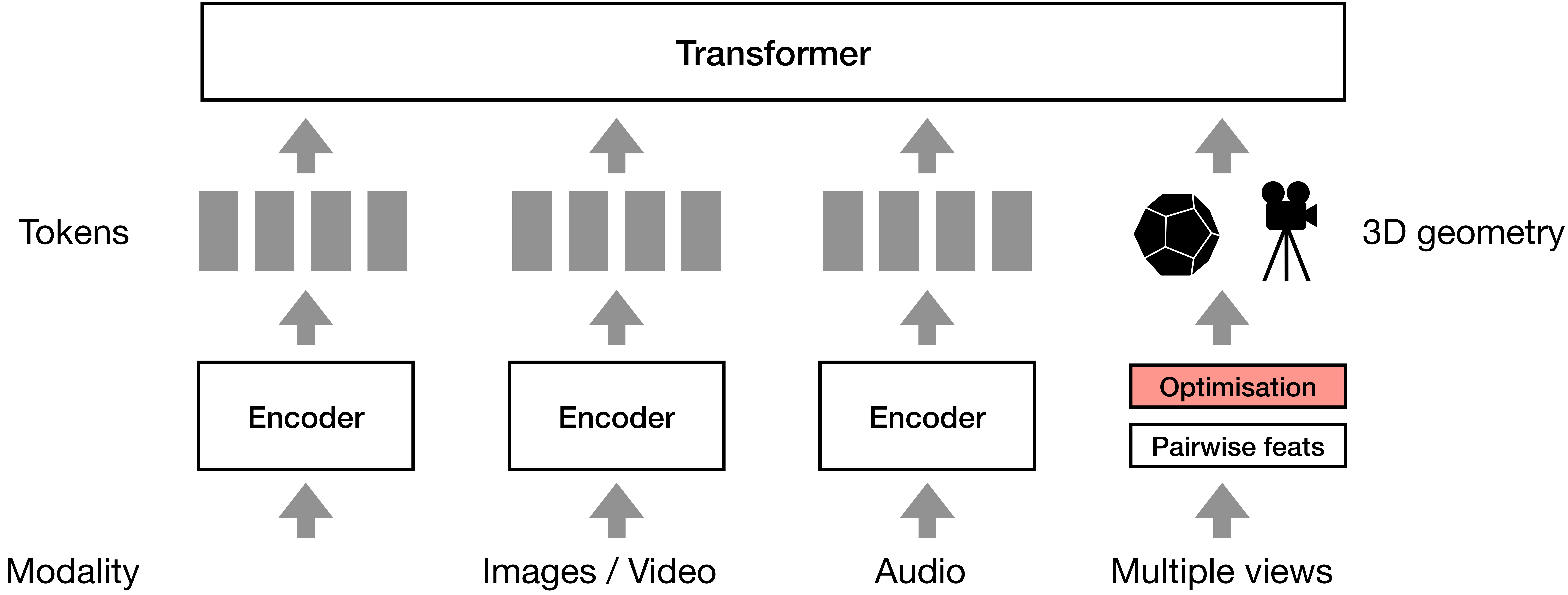
- What was the key ingredient in multi-view transformers?
  - Structure from motion was there since 1990s
  - Vision transformers appeared in 2019
  - Adapted to depth estimation in 2021
  - Point maps were introduced to the task in 2023
- Point maps: $I(u, v) \to (x, y, z)$
- Dynamic point maps: $I(u, v, t) \to (x, y, z, t)$

Video Dense Point Maps

- We will not go into details
- If you are curious, see

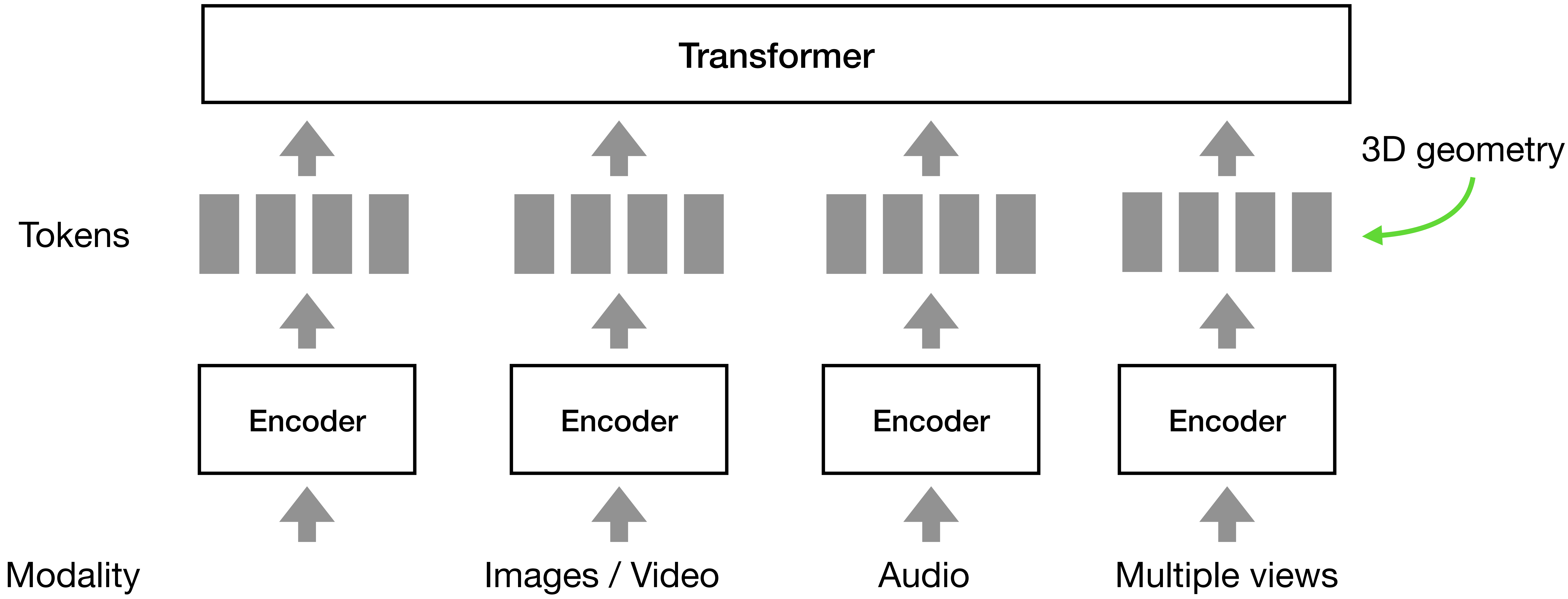


Foundational Models in Different Domains



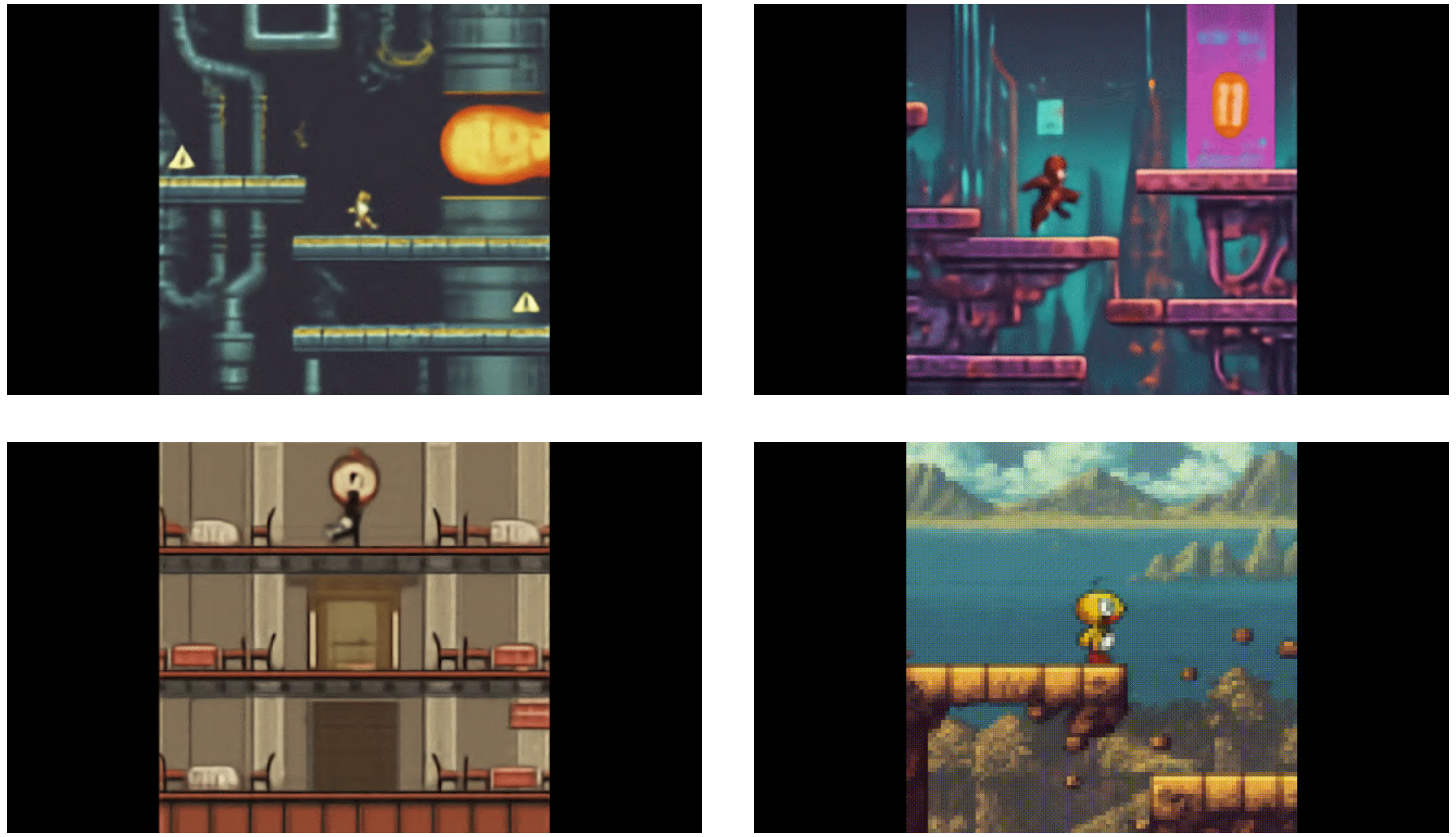
Slide adapted from “Where Geometry Goes” presentation by Andrea Vedaldi

Foundational Models in Different Domains



World Models

Genie



Genie 3

- Future entertainment & RL agent training environments
- Conditional video generation, coherent for ~60 seconds
- 3D representations can further improve coherence

Next Time: ML for Point Clouds